

PRATEEK SINGH

Gen AI | Node.js & Full-Stack Developer | Penetration Tester
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SUMMARY

Experienced in building Gen AI applications, RAG pipelines, RL-based security tools, and full-stack web systems (Node.js/Express, React, Python). Proficient in Python, LLMs, PyTorch, and penetration testing — actively practising on HackTheBox, TryHackMe, and Root-Me.

EDUCATION

B.Tech – Computer Science & Engineering (AI & ML)
Sharda University, Greater Noida | CGPA: 8.43

2022 – 2026

INTERNSHIP

Full-Stack Developer Intern | Uplyift

Jan 2026 – March 2026 | On-site, Delhi

- Building a Shopify CMS (React + FastAPI) with Gen AI (LLM-based) pipelines for automated SEO tag and product content generation, significantly reducing manual listing effort.
- Implemented Shopify Bulk Operations API with SSE streaming for real-time sync of thousands of variants; built Excel data pipeline (SheetJS + pandas + openpyxl) for bulk import and automated Shopify updates.
- Delivered end-to-end features in an agile team — REST API design, React UI components, and GraphQL integration with the Shopify Admin API.

PROJECTS

AI Resume Analyzer & Job Match Platform | *React, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, local LLM inference, Multer, pdf-parse*

- Building a full-stack platform with a Node.js/Express REST API backend, JWT authentication, and PostgreSQL (Prisma ORM) for managing users, resumes, and analysis history.
- Integrating Ollama directly into the Node.js backend for on-device LLM inference — no external API calls — to generate ATS scores, extract skills, and match resumes against job descriptions.
- Implementing resume parsing (PDF/DOCX via Multer + pdf-parse), skill-gap analysis, resume improvement suggestions.

AI Interview Preparation Platform | *React, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, local LLM inference*

- Architecting a Node.js/Express backend with JWT auth and a normalized PostgreSQL schema (Users → Interview Sessions → Questions → Answers → Feedback) via Prisma ORM.
- Integrating Ollama for local, offline interview-question generation by role (Frontend, Backend, Data Science) and AI evaluation of submitted answers.
- Designing scoring/feedback logic, weak-topic identification, and an interview history dashboard.

AI-Powered OWASP Top 10 Vulnerability Scanner | *Python, PyTorch (PPO + Self-Attention), Django, React*

- Hybrid scanner covering all OWASP Top 10 (2021): SQLi, XSS, NoSQLi, CMDi, Cryptographic Failures, Insecure Design, Security Misconfiguration, SSRF & more; traditional module achieved 92.4% precision with 4.3% false-positive rate.
- Engineered a PPO + Self-Attention (Transformer) RL agent for adaptive payload generation — 99% novelty (NoSQLi), 83.5% (CMDi), 67% (SQLi). React + Django stack with auto-generated PDF vulnerability reports.
- Validated on DVWA and OWASP Juice Shop.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++

Frontend: React.js, HTML, CSS

Backend: Node.js, Express.js, Django, FastAPI, REST APIs, Prisma ORM, JWT Authentication, Ollama (local LLM integration)

Databases & DevOps: PostgreSQL, MySQL, Docker, Git

AI / ML: LLMs, RAG Pipelines, PPO / RL, PyTorch, Scikit-learn, Pandas, NumPy

Security: Penetration Testing, Burp Suite, Metasploit, Nmap, OWASP ZAP, Scapy, Kali Linux, OWASP Top 10

CERTIFICATIONS & ACHIEVEMENTS

Udemy: [Learn Ethical Hacking from Scratch](#) (Aug 2023), [Learn Python and Ethical Hacking from Scratch](#) (Dec 2024)

TryHackMe: [Web Security Fundamentals](#) (Aug 2024)

Smart India Hackathon (SIH): Participant (Sep 2023)

DSA: 400+ problems solved on [LeetCode](#) & Codeforces